Lab 09: Launch RDS MySQL with Multi-AZ

In this lab, we will deploy an Amazon RDS MySQL database with Multi-AZ configuration for high availability and automatic failover.

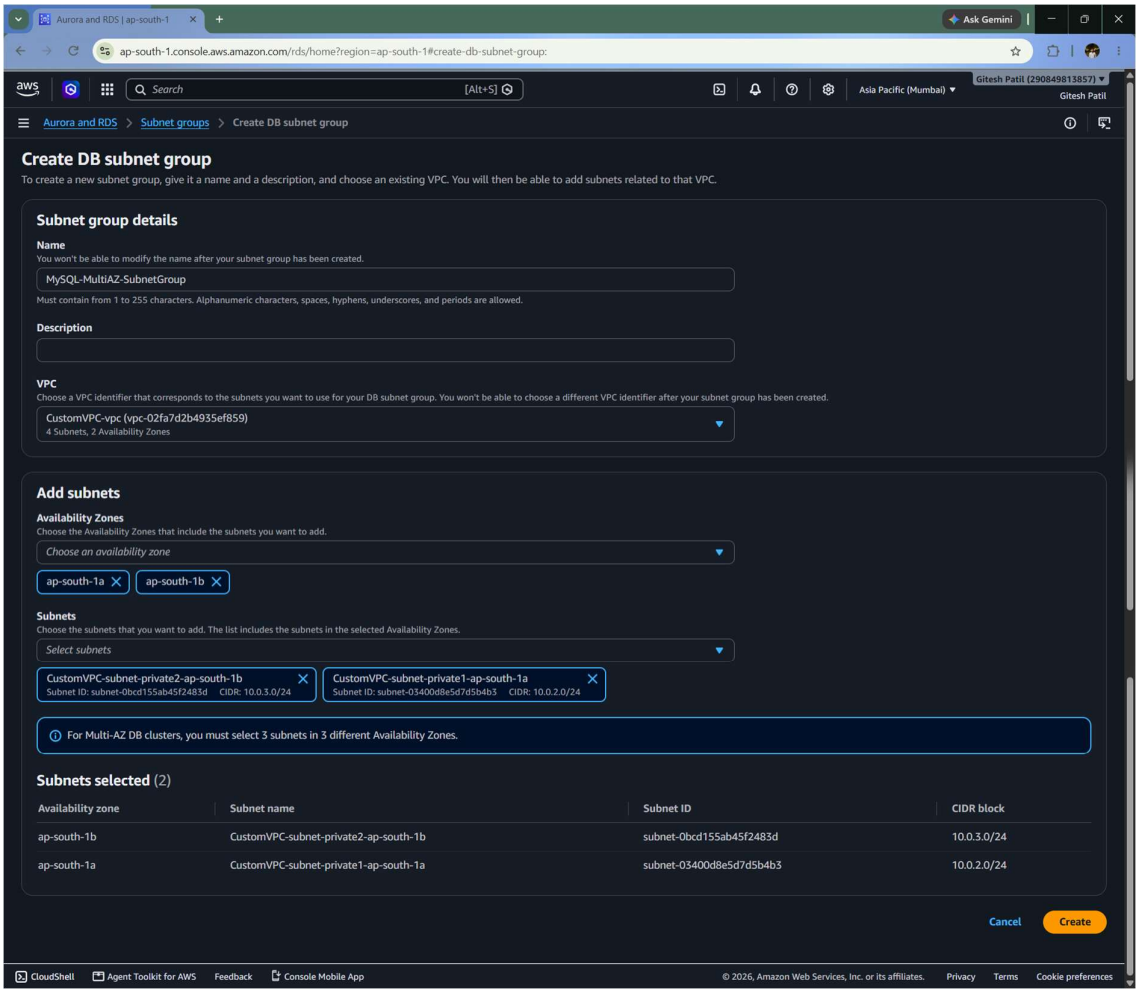
Prerequisites

- AWS Account with RDS permissions.
- VPC with at least two private subnets in different AZs.

STEPS

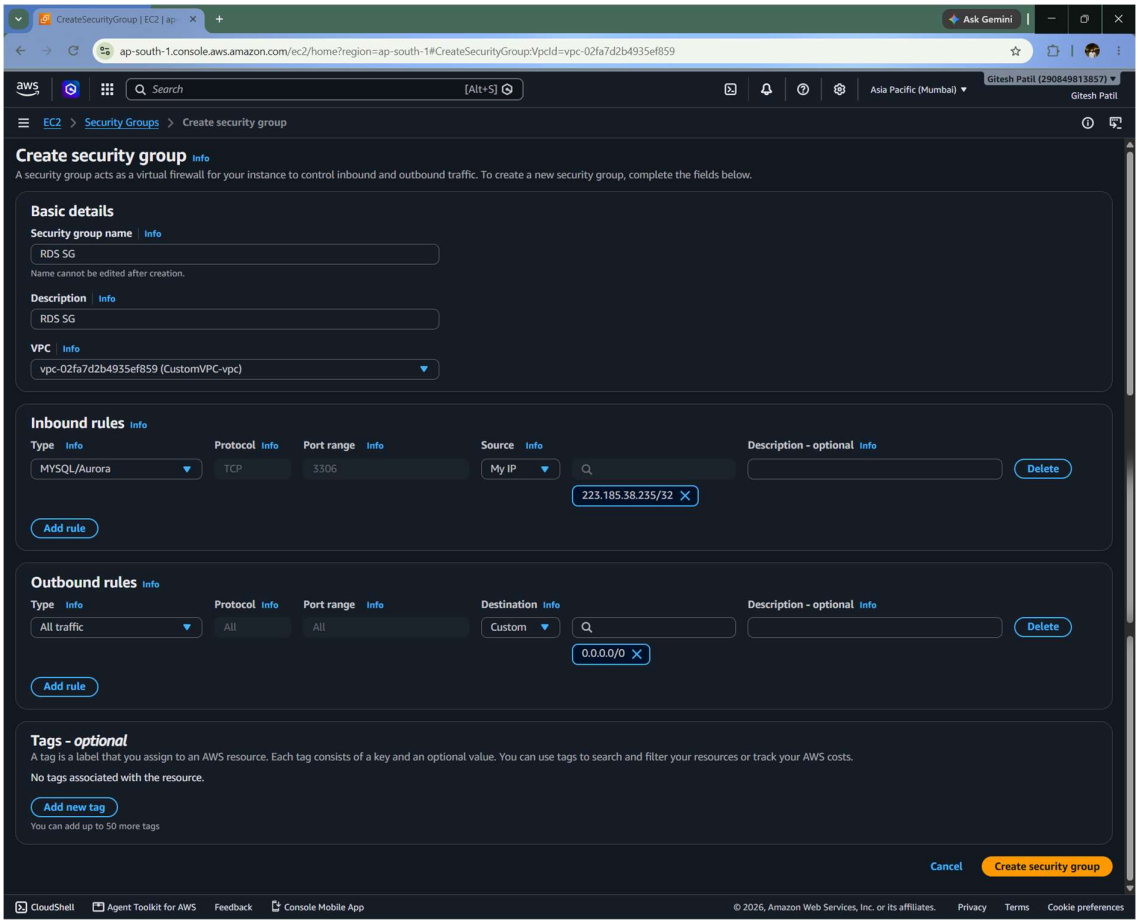
Step 1: Create DB Subnet Group

1. Go to RDS -> Subnet Groups Create DB Subnet Group.
2. Configure & create:
   - Name: MySQL-MultiAZ-SubnetGroup.
   - VPC: Select your VPC.
   - AZS : Select 2 different AZS
   - Subnets: Add two private subnets in different AZs.



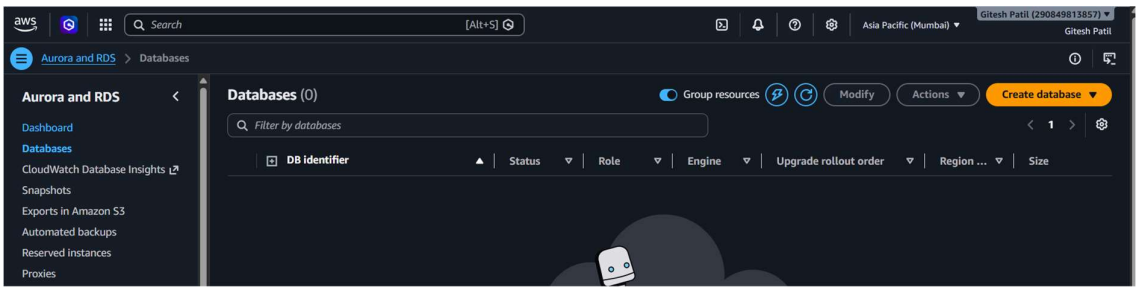
Step 2: Creating Security Groups

1. Navigate to EC2 Console and then from the left pane click on Security Groups.
2. Click on Create security group and configure as follows:
   - Security group name: RDS SG
   - Description: RDS SG
   - VPC: Select your VPC
   - Inbound Rule: Type -> MySQL/Aurora, Source Type -> My IP
3. Click on Create security group button and your SG will be created.



Step 3: Launch RDS MySQL Instance

1. Go to RDS Databases -> Create Database.



2. Select:

- Engine: MySQL.
- Version: Latest stable
- Template: Dev/Test
- Availablity & Durability: Multi-AZ DB instance deployment (2 instances)

3.Settings:

- DB Identifier: MySQL-MultiAZ.
- Master Username: admin.
- Password: Choose or auto-generate.

4.Instance Configuration:

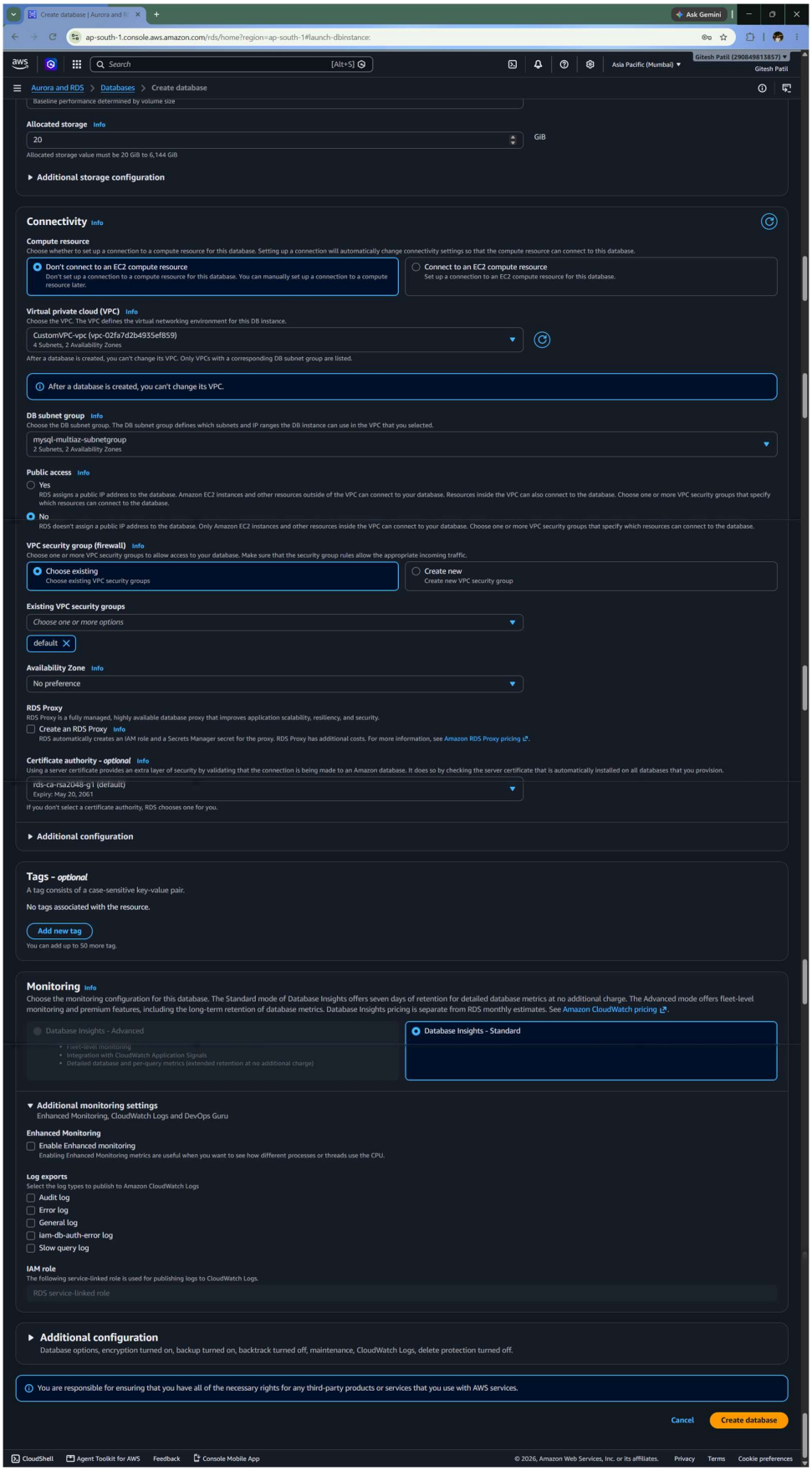
- DB Instance Class: db.t3.micro (for lab).
- Storage: General Purpose SSD.

5.Connectivity:

- VPC: Select your VPC.
- Subnet Group: Select the one created earlier.
- Public Access: No (recommended).
- Security Group: Select the one created earlier.

6. Uncheck Enable Enhanced Monitoring and then click Create Database.

7. Finally, you can go into the details and validate the instances.


Create database | Aurora and R
Ask Gemini
ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:
Search
[Alt+S]
Asia Pacific (Mumbai)
Gitesh Patil (290849813857)
Gitesh Patil
Aurora and RDS
Databases
Create database
Baseline performance determined by volume size
Allocated storage Info
20
GiB
Allocated storage value must be 20 GiB to 6,144 GiB
Additional storage configuration
Connectivity Info
Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.
Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.
Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.
CustomVPC-vpc (vpc-02fa7d2b4935ef859)
4 Subnets, 2 Availability Zones
After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.
After a database is created, you can't change its VPC.
DB subnet group Info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.
mysql-multiaz-subnetgroup
2 Subnets, 2 Availability Zones
Public access Info
Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.
VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
Choose existing
Choose existing VPC security groups
Create new
Create new VPC security group
Existing VPC security groups
Choose one or more options
default
Availability Zone Info
No preference
RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
Create an RDS Proxy Info
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing.
Certificate authority - optional Info
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061
If you don't select a certificate authority, RDS chooses one for you.
Additional configuration
Tags - optional
A tag consists of a case-sensitive key-value pair.
No tags associated with the resource.
Add new tag
You can add up to 50 more tag.
Monitoring Info
Choose the monitoring configuration for this database. The Standard mode of Database Insights offers seven days of retention for detailed database metrics at no additional charge. The Advanced mode offers fleet-level monitoring and premium features, including the long-term retention of database metrics. Database Insights pricing is separate from RDS monthly estimates. See Amazon CloudWatch pricing.
Database Insights - Advanced
Fleet-level monitoring
Integration with CloudWatch Application Signals
Detailed database and per-query metrics (extended retention at no additional charge)
Database Insights - Standard
Additional monitoring settings
Enhanced Monitoring, CloudWatch Logs and DevOps Guru
Enhanced Monitoring
Enable Enhanced monitoring
Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.
Log exports
Select the log types to publish to Amazon CloudWatch Logs
Audit log
Error log
General log
iam-db-auth-error log
Slow query log
IAM role
The following service-linked role is used for publishing logs to CloudWatch Logs.
RDS service-linked role
Additional configuration
Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.
You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.
Cancel
Create database
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

Create database | Aurora and R

ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

Aurora and RDS > Databases > Create database

Create database Info

Free plan has access to limited features and resources
The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. Learn more

Upgrade plan

Engine options

Engine type Info

- Aurora (MySQL Compatible)
- Aurora (PostgreSQL Compatible)
- MySQL
- PostgreSQL
- MariaDB
- Oracle
- Microsoft SQL Server
- IBM Db2

Choose a database creation method

- **Full configuration** — You set all of the configuration options, including ones for availability, security, backups, and maintenance.
- **Easy create** — Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates

Choose a sample template to meet your use case.

- Production — Use defaults for high availability and fast, consistent performance.
- Dev/Test — This instance is intended for development use outside of a production environment.
- **Free tier** — Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Availability and durability

Deployment options Info

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the Amazon RDS service level agreement (SLA).

- Multi-AZ DB cluster deployment (3 instances)
  Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
  - 99.95% uptime
  - Redundancy across Availability Zones
  - Increased read capacity
  - Reduced write latency

  Write/read endpoint — AZ 1: Primary instance + SSD
  Reader endpoints — AZ 2: Readable standby + SSD; AZ 3: Readable standby + SSD

- Multi-AZ DB instance deployment (2 instances)
  Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
  - 99.95% uptime
  - Redundancy across Availability Zones

  Write/read endpoint — AZ 1: Primary instance
  Standby (no endpoint) — AZ 2: Standby

- **Single-AZ DB instance deployment (1 instance)**
  Creates a single DB instance without standby instances. This setup provides:
  - 99.5% uptime
  - No data redundancy

  Write/read endpoint — AZ 1: Primary instance

Settings

Edition
MySQL Community

Engine version Info
View the engine versions that support the following database features.

Hide filters

- Show only versions that support the Multi-AZ DB cluster Info
  Create a A Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.
- Show only versions that support the Amazon RDS Optimized Writes Info
  Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version
MySQL 8.4.9

- Enable RDS Extended Support Info
  Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the RDS for MySQL documentation.

DB instance identifier Info
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
MySQL-MultiAZ
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username Info
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

- Managed in AWS Secrets Manager - *most secure*
  RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
- **Self managed**
  Create your own password or have RDS create a password that you manage.

- Auto generate password
  Amazon RDS can generate a password for you, or you can specify your own password.

You can view your credentials after you create your database. Click the 'View credential details' in the database creation banner to view the password.

Additional credentials settings

Database authentication options Info

- Password authentication
  Authenticates using database passwords.
- Password and IAM database authentication
  Authenticates using the database password and user credentials through AWS IAM users and roles.
- Password and Kerberos authentication
  Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class Info

Hide filters

- Show instance classes that support Amazon RDS Optimized Writes Info
  Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
- Include previous generation classes

- Standard classes (includes m classes)
- Memory optimized classes (includes r and x classes)
- Burstable classes (includes t classes)

Instance type
db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage Info
20 GiB
Allocated storage value must be 20 GiB to 6,144 GiB

Additional storage configuration

Connectivity Info

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

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